Modeling Student Work-Group Compatibility from Continuous Psychometric Profiles

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Abstract

Designing effective student project teams requires balancing interpersonal ease with complementary working styles. Self-selected groups tend to cluster similar personalities together, whereas random assignments often expose teams to coordination friction. Using primary survey data collected within our Albert School cohort ($N = 25$ complete responses out of 33 enrolled students on the five continuous scales of the *16Personalities* assessment, anonymized as `S01`–`S25`), this study develops a graph model of pairwise work-group compatibility. Rather than reducing individuals to discrete four-letter acronyms, we represent students as points in $[0, 1]^5$ and compare L_1 trait similarity (G_{sim}), equal-sparsity Euclidean similarity ($G_{\text{sim,4NN}}$), and pole opposition (G_{contrast}) against a domain-motivated compatibility network (G_{compat}) built from normalized L_2 proximity to an ideal partner profile and undirected k-nearest-neighbor selection ($k = 4$). Analyzing these networks in `NetworkX`, we evaluate connectivity sweeps, topological, raw, and mean-normalized ($\tilde{d} = (1-w)/\mu_{1-w}$) shortest paths, seven centrality series, Monte Carlo modularity null models, categorical alignment (ARI/NMI), and hyperparameter sensitivity. Our compatibility network (25 nodes, 73 edges, density 0.243) achieves targeted Energy disassortativity ($r_E = -0.463$) alongside cognitive alignment ($r_N = +0.448, r_T = +0.638$), remains compact (hop diameter 4, radius 2), and partitions the cohort into four collaboration profiles ($Q = 0.340$; ARI $= 0.376$ vs. four-letter types). Finally, Edmonds' maximum-weight matching incurs no loss in 12-dyad optimality on this cohort instance relative to K_{25} (88.0% mean pair fit), and our two-stage triad attachment (11 dyads + 1 triad) achieves +8.3 percentage points stronger triad cohesion (84.5% vs. 76.2%) than exact joint mean maximization at a cost of just 0.17 percentage points in cohort-wide mean fit.

1 Introduction

Collaborative project work plays a central role in applied data science and management education, yet assembling balanced student teams remains a delicate design problem. Experience in group coursework points to a recurring trade-off between similarity and complementarity. When students share similar ways of reasoning and communicating, day-to-day coordination is smoother, though uniform groups can suffer from blind spots when everyone approaches a problem from the same angle. Conversely, combining contrasting personalities broadens a team's perspective, but sharp differences in planning discipline or sensitivity to deadline stress frequently generate friction.

Network science provides a rigorous framework for studying such relational structures even when the underlying observations begin as tabular individual attributes [1]. In popular usage, personality instruments such as the *16Personalities* (NERIS Type Explorer) inventory [2] classify respondents into sixteen four-letter categories (for example, `INTJ` or `ENFP`). From a quantitative modeling standpoint,

however, binarizing continuous trait percentages at the 50% threshold discards substantial information. Two students scoring 49% and 51% on Extraversion are assigned opposite categorical labels despite nearly identical behavior, whereas two students scoring 51% and 100% share the same letter while exhibiting very different social dynamics.

Preserving the continuous geometry of the five trait scales allows us to formulate a testable research question: *Does modeling continuous psychometric coordinates as a target-partner compatibility graph capture relational structure, complementary pairing dynamics, and collaboration profiles beyond discrete four-letter MBTI categories, while supporting globally optimal team allocation?* To answer this question, we compare homophily and contrast baselines (including a density- and metric-matched 4-NN similarity benchmark) against our compatibility network, test modularity against topological and spatial null models, quantify partition divergence from categorical labels via Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI), and benchmark both hyperparameter sensitivity and odd-cohort triad resolution.

2 Dataset Presentation

2.1 Cohort Survey, Data Hygiene, and Trait Distributions

The dataset derives from an internal survey conducted within our Albert School cohort during the Autumn 2026 term on a shared Notion database. Classmates completed the standardized *16Personalities* assessment [2] and reported their categorical type, variant suffix (`-A` or `-T`), and exact continuous percentages across five dimensions rescaled to $[0, 1]$. Specifically, *Energy* (E) measures Extraverted initiative (> 0.5) against Introverted focus (< 0.5); *Mind* (N) contrasts Intuitive conceptual abstraction against Observant attention to concrete facts; *Nature* (T) distinguishes Thinking analytical evaluation from Feeling interpersonal prioritization; *Tactics* (J) captures Judging preference for structured planning over Prospecting adaptability; and *Identity* (A) reflects Assertive composure under pressure relative to Turbulent self-criticism.

Of the 33 students enrolled on the class roster, $N = 25$ reported complete numerical scores across all five dimensions and form the vertex set V of our networks. Prior to graph construction, consistency checks between declared four-letter types and continuous scores identified two entries in which respondents had transcribed the complementary pole percentage ($100\% - p$) on a single axis; both values were standardized to match the oriented scale convention (E, N, T, J, A). The 8 incomplete records lacked continuous trait percentages and were excluded rather than imputed, since substituting synthetic mean coordinates would artificially inflate central homophily. To protect classmate privacy both in this report and on our public GitHub repository, all respondent identifiers were randomly permuted and replaced with non-alphabetical anonymous codes (`S01`–`S25` for completed profiles, `S26`–`S33` for incomplete records) in `mbti_profiles_anon.csv`, while raw nominative files remain excluded via `.gitignore`.

Table 1: Summary statistics of the five continuous *16Personalities* scales across the $N = 25$ respondents.

Dimension	Pole Above 50%	Mean	Std. Dev.	Median	Min	Max
Energy (E)	Extraverted	53.8%	16.0%	53.0%	18.0%	100.0%
Mind (N)	Intuitive	60.7%	12.8%	63.0%	35.0%	86.0%
Nature (T)	Thinking	63.1%	18.0%	67.0%	22.0%	99.0%
Tactics (J)	Judging	57.9%	18.8%	57.0%	22.0%	92.0%
Identity (A)	Assertive	56.7%	22.1%	53.0%	12.0%	99.0%

Table 1 summarizes the empirical distribution of the five traits. Consistent with a quantitative business and data curriculum, the cohort leans toward Intuitive ($\bar{N} = 60.7\%$) and Thinking ($\bar{T} = 63.1\%$) orientations. Across the 11 categorical types observed in the sample, 15 students belong to the Analyst group (`ENTJ`: 7, `INTJ`: 4, `ENTP`: 3, `INTP`: 1), complemented by 5 Diplomats (`ENFP`: 3, `ENFJ`: 1, `INFJ`: 1), 4 Sentinels (`ESTJ`: 2, `ESFJ`: 1, `ISFJ`: 1), and 1 Explorer (`ISFP`: 1), with Assertive (14) and Turbulent (11)

variants well balanced. At the same time, Tactics (SD = 18.8%) and Identity (SD = 22.1%) span nearly the entire $[0, 1]$ interval, confirming wide continuous variation in organization and stress sensitivity that discrete type labels alone fail to capture.

2.2 Graph Formulation and Candidate Topologies

All networks in this study are defined as **simple, undirected, positively weighted graphs** $G = (V, E, w)$ on the fixed vertex set V ($|V| = 25$), where each vertex $u \in V$ carries its trait vector $x_u = (E_u, N_u, T_u, J_u, A_u) \in [0, 1]^5$ and categorical metadata. Each pair $\{u, v\}$ constitutes at most one prospective dyad with weight $w_{uv} \in [0, 1]$, and self-loops are excluded. Although k-nearest-neighbor rankings are inherently directed, collaboration requires bilateral commitment; we therefore symmetrize pairwise scores and take the **undirected k-NN union** rather than the mutual k-NN intersection. Empirically, restricting our $k = 4$ compatibility graph to mutual 4-NN retains only 27 edges and fragments the cohort into 5 components, leaving four peripheral students (`S05`, `S08`, `S10`, `S24`) isolated.

We compare our compatibility formulation against two primary baselines plus an equal-sparsity Euclidean benchmark on the same 25 vertices (Table 2). **Baseline 1 (G_{sim})** connects pairs with L_1 similarity $w_{uv}^{\text{sim}} = 1 - \frac{1}{5}\sum_{d=1}^{5} |x_{u,d} - x_{v,d}| \geq 0.80$ (20-point average gap), yielding 174 of $\binom{25}{2} = 300$ edges (density 0.580). **Baseline 2 (G_{contrast})** measures pole opposition across 0.50 via $w_{uv}^{\text{con}} = \frac{4}{5}\sum_{d=1}^{5} \max\big(0, -(x_{u,d} - 0.5)(x_{v,d} - 0.5)\big) \in [0, 1]$ and takes the top-5 union (97 edges, density 0.323). Because comparing G_{sim} (174 edges, L_1 norm) directly to a 4-NN L_2 graph confounds density and norm choice with the compatibility rule itself, we also evaluate a controlled benchmark $G_{\text{sim,4NN}}$ using normalized Euclidean similarity $w_{uv}^{\text{euc}} = 1 - \|x_u - x_v\|_2/\sqrt{5}$ and undirected 4-NN selection (71 edges, density 0.237).

Our final model, **Work-Group Compatibility (G_{compat})**, maps each student x_u to an ideal partner vector $x_u^* \in [0, 1]^5$:

$$x_u^* = \left(1 - E_u,\ N_u,\ T_u,\ \frac{J_u + \alpha}{2},\ \frac{A_u + \alpha}{2}\right), \qquad \alpha = 0.65. \tag{1}$$

Equation (1) combines complementary social energy ($1 - E_u$), aligned perception and communication (N_u, T_u), and moderated planning and composure targets pulled halfway toward $\alpha = 0.65$ (around the 60th–65th cohort percentile, slightly above means $\bar{J} = 57.9\%, \bar{A} = 56.7\%$). Whereas G_{sim} uses L_1 for direct percentage-point thresholds, here we use **normalized Euclidean distance** $d(u \to v) = \|x_v - x_u^*\|_2/\sqrt{5} \in [0, 1]$ so severe single-trait clashes are penalized quadratically rather than masked by four close traits:

$$w_{uv} = 1 - \frac{\|x_v - x_u^*\|_2 + \|x_u - x_v^*\|_2}{2\sqrt{5}} \in [0, 1]. \tag{2}$$

Undirected 4-NN selection yields a connected 73-edge network (density 0.243, degrees in $[4, 11]$).

Table 2: Structural summary of the baseline, controlled 4-NN similarity, and compatibility models ($|V| = 25$).

Graph Model	Construction Rule	Edges	Density	Deg. Range	Weight Range	Connected
G_{sim} (Baseline 1)	L_1 similarity ≥ 0.80	174	0.580	$[3, 20]$	$[0.800, 0.948]$	Yes (1 comp.)
G_{contrast} (Baseline 2)	Top-5 pole-opposition union	97	0.323	$[5, 15]$	$[0.024, 0.248]$	Yes (1 comp.)
$G_{\text{sim,4NN}}$ (Controlled Sim.)	$L_2/\sqrt{5}$ similarity + 4-NN	71	0.237	$[4, 9]$	$[0.739, 0.936]$	Yes (1 comp.)
G_{compat} (Final Model)	$L_2/\sqrt{5}$ target fit ($\alpha = 0.65$) + 4-NN	**73**	**0.243**	$[\mathbf{4}, \mathbf{11}]$	$[\mathbf{0.758}, \mathbf{0.944}]$	**Yes (1 comp.)**

3 Description of the Analysis

3.1 Structural Metrics, Null Models, and Validation Pipeline

First, we sweep each sparsity control (G_{sim} cutoffs in $[0.60, 0.95]$; neighbor counts $k \in \{1, \ldots, 24\}$ for contrast and compatibility) to track component fragmentation and isolates. Within each connected component $C \subseteq V$, node eccentricity $\epsilon(u) = \max_{v \in C} d(u, v)$ defines diameter $D(C)$, radius $R(C)$, center $\{u : \epsilon(u) = R(C)\}$, and periphery $\{u : \epsilon(u) = D(C)\}$ ($d(u, v) = \infty$ across disconnected components). We

evaluate three shortest-path conventions: (i) unweighted hop counts, (ii) raw Dijkstra friction cost $d_{uv} = 1 - w_{uv}$, and (iii) **mean-normalized friction** $\tilde{d}_{uv} = (1 - w_{uv})/\mu_{1-w}$, where $\mu_{1-w} = \frac{1}{|E|}\sum_{e \in E}(1 - w_e)$ rescales each graph so its average edge has unit cost ($\mathbb{E}[\tilde{d}_e] = 1$), eliminating scale distortion when comparing low-weight G_{contrast} ($\mu_{1-w} = 0.923$) against high-weight proximity models ($\mu_{1-w} \approx 0.14$–0.15).

Second, we evaluate seven centrality series [3, 4, 5]: unweighted degree, weighted strength $s_u = \sum_{v \in \mathcal{N}(u)} w_{uv}$, unweighted (hop) and friction-weighted ($d_{uv} = 1 - w_{uv}$) betweenness, unweighted and friction-weighted Wasserman–Faust closeness, and component-wise weighted eigenvector centrality. Because Equation (1) pulls ideal partners toward $(0.50, N_u, T_u, 0.65, 0.65)$, central trait profiles naturally attract more links; we quantify this construction effect via Spearman rank correlations (ρ_s) between Euclidean proximity to the cohort centroid $\bar{x}$ and centrality, alongside edge-weighted continuous trait assortativities r_d ($d \in \{E, N, T, J, A\}$).

Third, we detect communities via weighted Louvain optimization [6], maximizing Newman–Girvan weighted modularity:

$$Q = \frac{1}{2m} \sum_{u,v \in V} \left[w_{uv} - \frac{s_u s_v}{2m} \right] \delta(c_u, c_v), \qquad 2m = \sum_{u,v \in V} w_{uv}. \tag{3}$$

We analyze modularity-plateau stability across 100 random seeds (0–99), benchmark against Clauset–Newman–Moore (CNM) greedy modularity [7] and unweighted Girvan–Newman splits [8] (evaluating all partitions under weighted Q on G_{compat}), test Q_{obs} against two 1,000-run Monte Carlo null models (degree-preserving topological rewiring and independent five-column trait permutation with 4-NN reconstruction), and compute ARI/NMI against categorical MBTI groups. Finally, we solve Edmonds' maximum-weight matching [9] (`nx.max_weight_matching`), benchmark our two-stage odd-cohort triad heuristic against exhaustive joint 1-triad + 11-dyad optimization across all $\binom{25}{3} = 2{,}300$ candidate trios, and sweep (k, α) across $k \in \{3, 4, 5\}$ and $\alpha \in [0.55, 0.75]$.

3.2 Implementation, Code Originality, and AI Usage

All analyses were implemented in Python ($\geq$ 3.11, managed with `uv`) using `NetworkX` [10], `NumPy`, `Pandas`, `SciPy`, `Plotly`, and `PyVis`. The anonymized dataset (`mbti_profiles_anon.csv`), notebook (`base_notebook.ipynb`), interactive explorer (`graph.html`), and validation script (`validation_checks.py`) are provided in the companion GitHub repository linked beneath the title block. In accordance with academic integrity rules, **no pre-existing code associated with this dataset was reused**; all modeling, optimization, and visualization code was written by the authors. Generative AI tools were used to audit methodological rigor (flagging density confounds, modularity null models, and friction-scale normalization), assist with package/export debugging, and polish LaTeX formatting; all modeling decisions, mathematical formulations, and interpretations were verified by the three authors.

4 Results

4.1 Network Topologies and Assortativity Verification

Visualizing the graphs illustrates why neither baseline yields balanced teams (Figures 1 and 2). Thresholded G_{sim} (174 edges, mean degree 13.92) connects central profiles to up to 20 peers while leaving `S05` and `S24` at degree 3. Conversely, G_{contrast} (97 edges) concentrates disassortative stars around extreme profiles such as `S24` (`ENTJ-A`: $E = 100\%, T = 86\%, A = 99\%$) and `S05` (`ENFP-T`: $N = 86\%, T = 22\%, A = 12\%$; 24.8% contrast), pairing opposing planning and stress habits.

By contrast, G_{compat} (Figure 2) forms a sparse 73-edge graph (mean degree 5.84, weights in $[75.8\%, 94.4\%]$, mean 85.9%). Edge-weighted continuous trait assortativities verify that the model behaves as designed relative to controlled Euclidean similarity $G_{\text{sim,4NN}}$: Extraversion assortativity flips from homophily on

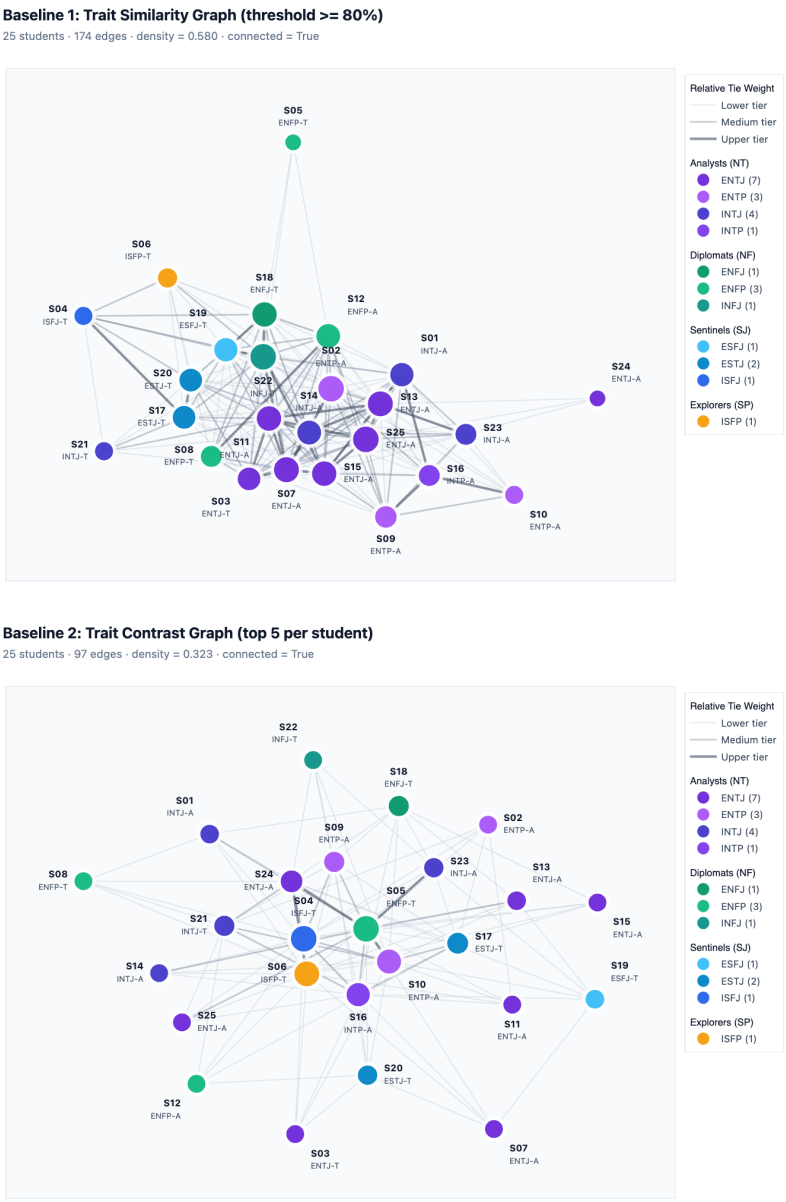


Figure 1: Baseline networks ($N = 25$). *Top:* Trait Similarity (G_{sim}, 174 edges, density 0.580). *Bottom:* Trait Contrast (G_{contrast}, 97 edges, density 0.323).

$G_{\text{sim,4NN}}$ ($r_E = +0.242$) to strong complementarity on G_{compat} ($r_E = -0.463$), while cognitive and organizational alignment remain strongly positive ($r_N = +0.448$, $r_T = +0.638$, $r_J = +0.541$, $r_A = +0.536$). Strong ties ($\geq 88\%$) thus join introverted and extraverted classmates with shared analytical framing, such as `S01` (`INTJ-A`, $E = 36\%$) with `S25` (`ENTJ-A`, $E = 67\%$, 94.4%), `S14` (`INTJ-A`, $E = 43\%$) with `S11` and `S13` (`ENTJ-A`, 93.8% each), and `S18` (`ENFJ-T`, $E = 56\%$) with `S22` (`INFJ-T`, $E = 46\%$, 92.2%).

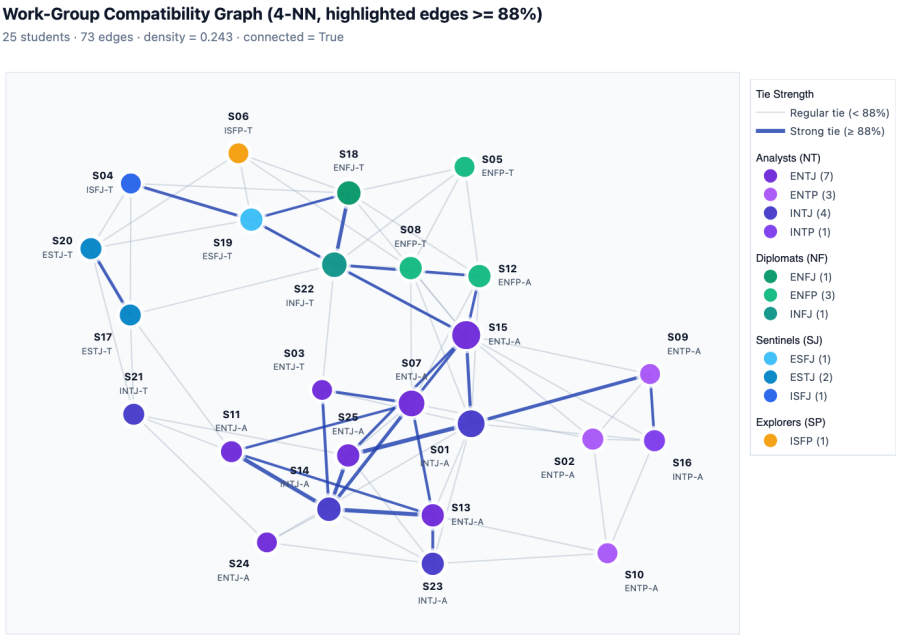


Figure 2: Work-Group Compatibility Network (G_{compat}, 4-NN, 25 nodes, 73 edges). Dark blue lines denote ties with compatibility $\geq 88\%$.

4.2 Connectivity Robustness and Mean-Normalized Shortest Paths

Parameter sweeps uncover sharp differences in sparsification resilience (Figure 3). Raising the G_{sim} cutoff from 0.80 to 0.85 (87 edges, still denser than G_{compat}) splits the cohort into 3 components and isolates `S05` and `S24`; at 0.90 (25 edges), it fragments into 7 components (4 isolates, 64.0% in the largest cluster). Meanwhile, G_{compat} is disconnected only at $k = 1$ (5 components, 32.0% largest) and reaches 100% single-component connectivity for all $k \geq 2$ (G_{contrast} is connected from $k \geq 1$).

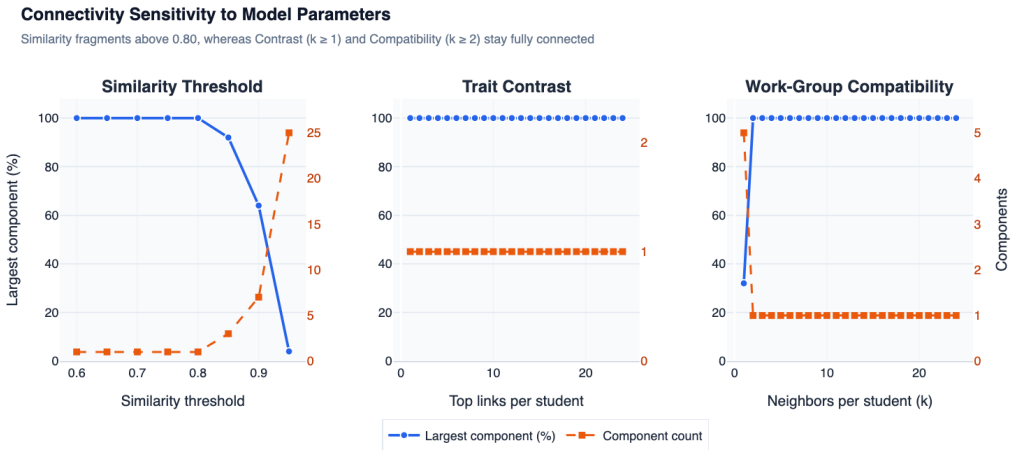


Figure 3: Connectivity sensitivity across parameter sweeps: share of students in the largest component (solid blue) and total component count (dashed orange).

Table 3 and Figure 4 compare topological, raw friction ($d = 1 - w$), and mean-normalized friction

($\tilde{d} = (1-w)/\mu_{1-w}$) shortest paths. Although G_{contrast} displays a raw cost diameter of 2.705 versus 0.619 on G_{compat}, this $\times 4$ gap is purely a **weight-ceiling effect**: contrast weights peak at 0.248, giving per-edge friction $\mu_{1-w} = 0.923$ versus 0.141 on G_{compat}. Rescaling by μ_{1-w} places all models in comparable average-edge units ($\tilde{D} = 3.58, 2.93, 4.60$, and 4.37). Comparing G_{compat} directly against scale-matched $G_{\text{sim,4NN}}$ ($\mu_{1-w} = 0.148$) shows that at virtually identical density, G_{compat} achieves a tighter hop radius ($R = 2$ vs. 3), shorter raw cost bounds (0.619/0.312 vs. 0.680/0.387), and smaller mean-scaled bounds (4.37/2.21 vs. 4.60/2.62), uniquely centered on `S15` with weighted periphery {`S05`, `S10`} (`S05` → `S22` → `S15` → `S16` → `S10`).

Table 3: Topological, raw friction $(1-w)$, and mean-normalized ($(1-w)/\mu_{1-w}$) shortest-path bounds ($n = 25$).

Graph Model	Edges	μ_{1-w}	Hop D/R	Raw D_w/R_w	Mean-Scaled $\tilde{D}/\tilde{R}$	W. Center	W. Periphery
G_{sim} ($L_1 \geq 0.80$)	174	0.146	3/2	0.522/0.278	3.58/1.91	`S15`	`S05, S24`
G_{contrast} (top-5)	97	0.923	3/2	2.705/1.746	2.93/1.89	`S24`	`S12, S19`
$G_{\text{sim,4NN}}$ (L_2 4-NN)	71	0.148	4/3	0.680/0.387	4.60/2.62	`S15`	`S05, S24`
G_{compat} (L_2 4-NN)	**73**	**0.141**	**4/2**	**0.619/0.312**	**4.37/2.21**	`S15`	`S05, S10`

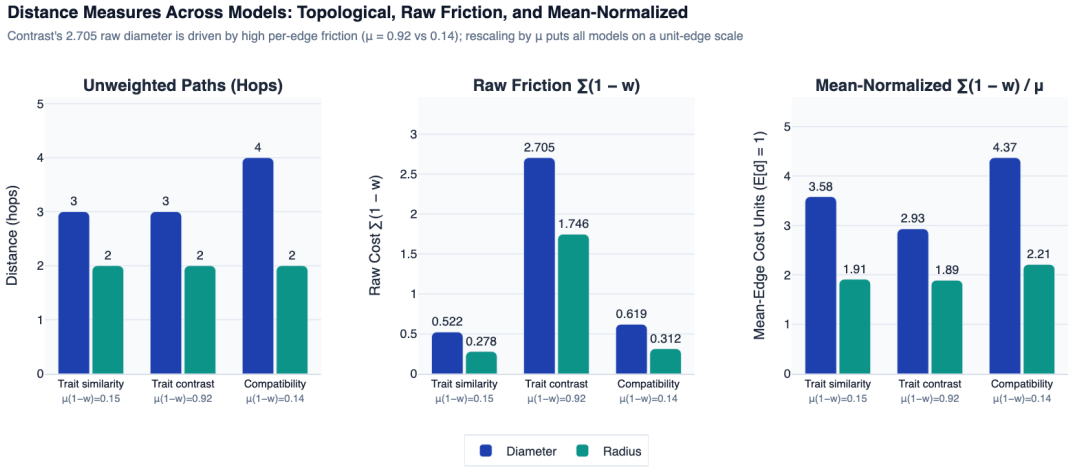


Figure 4: Diameter and radius across models under unweighted hops, raw friction $\sum(1-w)$, and mean-normalized friction $\sum(1-w)/\mu_{1-w}$.

4.3 Centrality Profiles and Geometric Proximity Analysis

In G_{sim}, high density compresses centrality differences (`S02`, `S22`, `S25` tie at degree 20; hop betweenness ≤ 0.067), while G_{contrast} turns extreme profiles into dominant hubs (`S04`, `S05` at degree 15; `S05` weighted betweenness 0.348). On G_{compat} (Table 4 and Figure 5), the **top three core hubs (`S15`, `S01`, `S07`) hold ranks #1–#3 across local, unweighted geodesic, weighted betweenness, and spectral centralities**, while `S22` (`INFJ-T`) places #4 across degree, strength, hop betweenness, hop closeness, and eigenvector centrality. Incorporating continuous tie friction ($d = 1 - w$) refines the geodesic rankings: `S19` (`ESFJ-T`, 0.105) ties `S22` for #4 in weighted betweenness, while high-affinity Analyst connectors `S25` (4.616, #3) and `S14` (4.499, #5) enter the Top 5 in weighted closeness ahead of `S22` (4.373).

Methodologically, this hub hierarchy should not be mistaken for a model-free empirical surprise: because Equation (1) mirrors E around 0.50 and pulls J, A toward 0.65, node centrality primarily reflects proximity to the center of the trait space. Spearman correlations between Euclidean proximity to the cohort centroid $\bar{x}$ and centrality confirm strong geometric coupling on weighted strength ($\rho_s = 0.799, p = 1.7 \times 10^{-6}$), eigenvector ($\rho_s = 0.720, p = 5.0 \times 10^{-5}$), and degree ($\rho_s = 0.717, p = 5.4 \times 10^{-5}$), explaining why balanced profiles (`S15`: $E = 52\%$; `S07`: 51%; `S22`: 46%) attract extra incoming links. Notably, hop betweenness correlates more moderately with centroid proximity ($\rho_s = 0.591, p = 1.9 \times 10^{-3}$), leaving room for asymmetric cohort effects: `S01` (`INTJ-A`, $E = 36\%$) ranks #2 overall because seven extraverted `ENTJ`s compete for a smaller pool of introverted `NT-J` counterparts.

Table 4: Key hubs and friction-weighted connectors in G_{compat} across all seven centrality series.

Student	MBTI	Deg.	Strength	Betw. (Hops)	W. Betw. $(1-w)$	Clos. (Hops)	W. Clos. $(1-w)$	Eigenvector
S15	ENTJ-A	**11**	**9.605**	**0.199**	**0.236**	**0.649**	**5.097**	**0.386**
S01	INTJ-A	10	8.657	0.144	0.149	0.585	4.761	0.323
S07	ENTJ-A	9	7.912	0.103	0.109	0.585	4.586	0.310
S22	INFJ-T	8	6.963	0.081	0.105	0.545	4.373	0.255
S14	INTJ-A	7	6.227	0.026	0.080	0.490	4.499	0.216
S25	ENTJ-A	6	5.332	0.045	0.072	0.533	4.616	0.240
S19	ESFJ-T	6	5.249	0.050	0.105	0.500	3.947	0.175



Figure 5: Top five students across local, spectral, hop-geodesic, and friction-weighted $(1-w)$ centralities on G_{compat}.

4.4 Community Structure, Modularity Plateau, and Null Models

Table 5 evaluates community structure across models, algorithms, seed stability, and null benchmarks. Comparing thresholded G_{sim} ($Q = 0.126$, density 0.580) with equal-sparsity $G_{\text{sim,4NN}}$ ($Q = 0.399$, density 0.237) confirms that G_{sim}'s weak modularity stems from excessive density rather than homophily, while G_{contrast} yields low modularity ($Q = 0.117$) because assortative modularity Q is structurally ill-suited to disassortative opposition graphs.

On G_{compat}, sweeping 100 random seeds (0–99) shows that 88/100 **runs converge to $k = 4$ communities** ($Q \in [0.334, 0.347]$, with $[9, 6, 6, 4]$ at `seed=42` as the plurality modal partition in 46/100 runs), while 12/100 runs merge down to $k = 3$ ($Q \in [0.322, 0.351]$, peaking at $Q_{\max} = 0.3505$ on $[10, 10, 5]$ in 5 runs). Because small graphs exhibit **modularity-plateau near-degeneracy** in which modularity's resolution scale slightly rewards merging adjacent clusters (+0.010 gain), we retain the plurality $k = 4$ mode ($Q = 0.340$), which outperforms both CNM greedy modularity ($Q = 0.306$) and unweighted Girvan–Newman at $k = 4$ (evaluated under weighted Q: 0.299, sizes $[7, 7, 7, 4]$) while preserving four interpretable archetypes instead of two coarse 10-student blocks.

Our 1,000-run Monte Carlo null tests further clarify what $Q = 0.340$ represents. Relative to degree-preserving topological rewiring ($Q_{\text{rewire}} = 0.261 \pm 0.021$), G_{compat} is strongly modular ($z = +3.75, p = 0.002$). Relative to independent five-column trait permutations reconstructed as 4-NN compatibility graphs ($Q_{\text{perm}} = 0.351 \pm 0.038, z = -0.29, p = 0.58$), Q_{obs} matches the spatial expectation: any 4-NN graph in $[0, 1]^5$ inherits $Q \approx 0.35$ from Euclidean triangle closure, while mirroring $E \mapsto 1 - E$ slightly lowers Q relative to $G_{\text{sim,4NN}}$ (0.340 vs. 0.399) by cross-linking introverts and extraverts. The clusters therefore delineate coherent geometric zones of mutually compatible profiles rather than multimodal gaps in raw trait marginals.

Table 5: Weighted modularity (Q) across models, algorithms, and 1,000-run Monte Carlo null benchmarks.

Model / Benchmark	Clusters (k)	Sizes / Summary	Weighted Q	Interpretation
G_{sim} ($L_1 \geq 0.80$, Louvain)	2	[14, 11]	0.126	Blurred by high density (0.580)
G_{contrast} (top-5, Louvain)	3	[9, 9, 7]	0.117	Low Q (Q ill-suited to disassortative ties)
$G_{\text{sim,4NN}}$ (L_2 4-NN, Louvain)	4	[9, 6, 5, 5]	0.399	Segregates E ($r_E = +0.242$)
G_{compat} (Louvain mode / `seed=42`)	**4**	**[9, 6, 6, 4]**	**0.340**	**Plurality mode** (46/100; $k = 4$ in 88/100)
G_{compat} (CNM Greedy Modularity)	3	[12, 7, 6]	0.306	Over-merges `C2` and `C3` into $n = 12$
G_{compat} (Girvan–Newman, $k = 4$ split)	4	[7, 7, 7, 4]	0.299	Unweighted edge-betweenness split
Null A: Topological rewiring (1,000×)	—	$\mu \pm \sigma$	0.261 ± 0.021	Significant vs. random topology ($z = +3.75, p = 0.002$)
Null B: Column-permuted 4-NN (1,000×)	—	$\mu \pm \sigma$	0.351 ± 0.038	Expected 5D 4-NN spatial locality ($z = -0.29, p = 0.58$)

Table 6: Structural metrics and mean trait profiles of the four Louvain communities on G_{compat}.

Community	n	Density	Mean Tie	$\bar{E}$	$\bar{N}$	$\bar{T}$	$\bar{J}$	$\bar{A}$
`C1`: Strategic Drivers	9	0.56	87.3%	58.4%	66.9%	74.6%	68.7%	69.7%
`C2`: Empathetic Diplomats	6	0.80	84.5%	58.3%	69.3%	43.8%	48.5%	42.7%
`C3`: Structured Organizers	6	0.60	85.8%	41.3%	43.8%	54.5%	70.0%	38.3%
`C4`: Assertive Analysts	4	0.83	84.9%	55.5%	59.2%	79.0%	29.8%	76.0%

Comparing these communities (Table 6, Figure 6) against discrete MBTI labels directly tests our research question: alignment is moderate against the 4 NERIS role groups (ARI = 0.429, NMI = 0.613), the 11 four-letter types (ARI = 0.376, NMI = 0.656), and the 13 type-variant classes (ARI = 0.297, NMI = 0.658). Continuous geometry adds substantive resolution: it separates the 15-student Analyst bloc into structured `NT-J` *Strategic Drivers* (`C1`, $n = 9$, $\bar{J} = 68.7\%$) versus exploratory `NT-P -A` *Assertive Analysts* (`C4`, $n = 4$, $\bar{J} = 29.8\%, \bar{A} = 76.0\%$), and assigns borderline `S15` (`ENTJ-A`, $E = 52\%, T = 54\%$) to *Empathetic Diplomats* (`C2`, $n = 6$, $\bar{N} = 69.3\%, \bar{T} = 43.8\%$). Meanwhile, *Structured Organizers* (`C3`, $n = 6$, $\bar{J} = 70.0\%, \bar{N} = 43.8\%$) combine introverted planners (`ISFJ-T`, `ISFP-T`, `INTJ-T`) with moderate-Extraversion Sentinels (`ESTJ-T` at $E = 51\%$–58%, `ESFJ-T` at 54%) around concrete, deadline-focused execution.

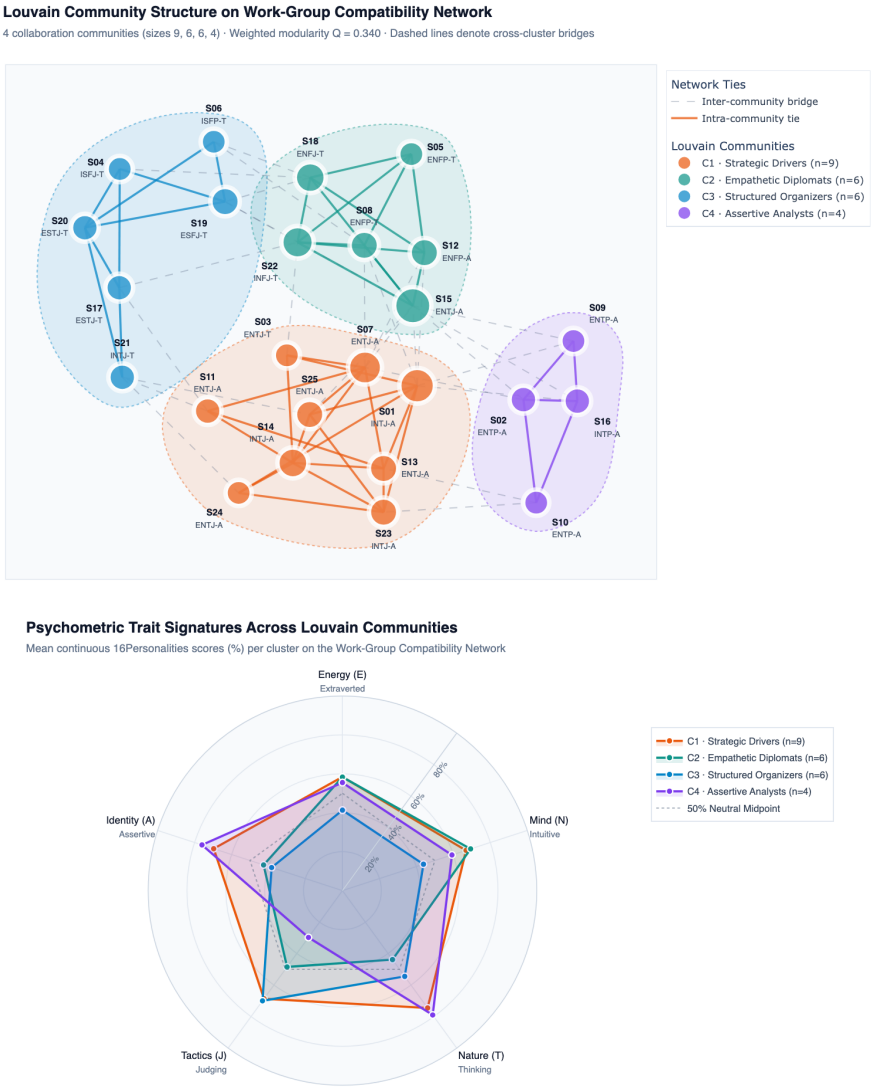


Figure 6: *Top:* Louvain community partition ($Q = 0.340$) on G_{compat}, showing intra-community links (solid) and cross-community bridges (dashed). *Bottom:* Mean five-trait radar profiles across the four communities.

4.5 Optimal Team Allocation, Triad Benchmark, and Sensitivity

Solving Edmonds' maximum-weight matching on G_{compat} yields 12 dyads with total weight 10.556 (88.0% mean compatibility, range 76.8%–94.4%; Table 7 and Figure 7). On this cohort instance, our 73-edge 4-NN graph achieves the exact same 10.556 12-dyad optimum as the complete graph K_{25} (300 edges), whereas matching on pure similarity $G_{\text{sim,4NN}}$ drops mean team compatibility to 85.2%. Because $N = 25$ is odd, we absorb unmatched student `S05` (`ENFP-T`) via a **two-stage post-matching heuristic**, fixing the 12 Edmonds dyads and attaching `S05` to the pair maximizing three-way mean fit (`{S05, S18, S22}` at 84.5%). Benchmarking against the **exact joint 1-triad + 11-dyad maximum-mean optimum** across all $\binom{25}{3} = 2{,}300$ candidate trios confirms that while exact joint optimization reaches 87.50% cohort-wide mean fit vs. 87.33% (+0.17 percentage points, by freeing `S18`–`S22` at 92.2% and forming trio `{S05, S06, S08}`), it degrades the trio itself to 76.2%. Our two-stage allocation thus delivers a **+8.3-point gain on the three-person team** and superior max-min fairness (76.8% vs. 76.2% weakest team). Ten of the 12 teams remain intra-cluster, reflecting internal structural consistency across methods sharing the same weights w_{uv}.

Table 7: Optimal project team assignment covering all $N = 25$ students (11 dyads and 1 triad).

Team	Format	Assigned Members (MBTI)	Cluster Alignment	Compatibility
Team 01	Dyad (2)	`S01` (`INTJ-A`), `S25` (`ENTJ-A`)	Intra-`C1`	94.4%
Team 02	Dyad (2)	`S11` (`ENTJ-A`), `S14` (`INTJ-A`)	Intra-`C1`	93.8%
Team 03	Dyad (2)	`S17` (`ESTJ-T`), `S20` (`ESTJ-T`)	Intra-`C3`	90.3%
Team 04	Dyad (2)	`S04` (`ISFJ-T`), `S19` (`ESFJ-T`)	Intra-`C3`	89.1%
Team 05	Dyad (2)	`S03` (`ENTJ-T`), `S07` (`ENTJ-A`)	Intra-`C1`	89.1%
Team 06	Dyad (2)	`S13` (`ENTJ-A`), `S23` (`INTJ-A`)	Intra-`C1`	89.0%
Team 07	Dyad (2)	`S09` (`ENTP-A`), `S16` (`INTP-A`)	Intra-`C4`	88.6%
Team 08	Dyad (2)	`S12` (`ENFP-A`), `S15` (`ENTJ-A`)	Intra-`C2`	88.6%
Team 09	Dyad (2)	`S02` (`ENTP-A`), `S10` (`ENTP-A`)	Intra-`C4`	81.9%
Team 10	Dyad (2)	`S06` (`ISFP-T`), `S08` (`ENFP-T`)	Cross (`C3` ↔ `C2`)	81.8%
Team 11	Dyad (2)	`S21` (`INTJ-T`), `S24` (`ENTJ-A`)	Cross (`C3` ↔ `C1`)	76.8%
Team 12	Triad (3)	`S05` (`ENFP-T`), `S18` (`ENFJ-T`), `S22` (`INFJ-T`)	Intra-`C2`	84.5%

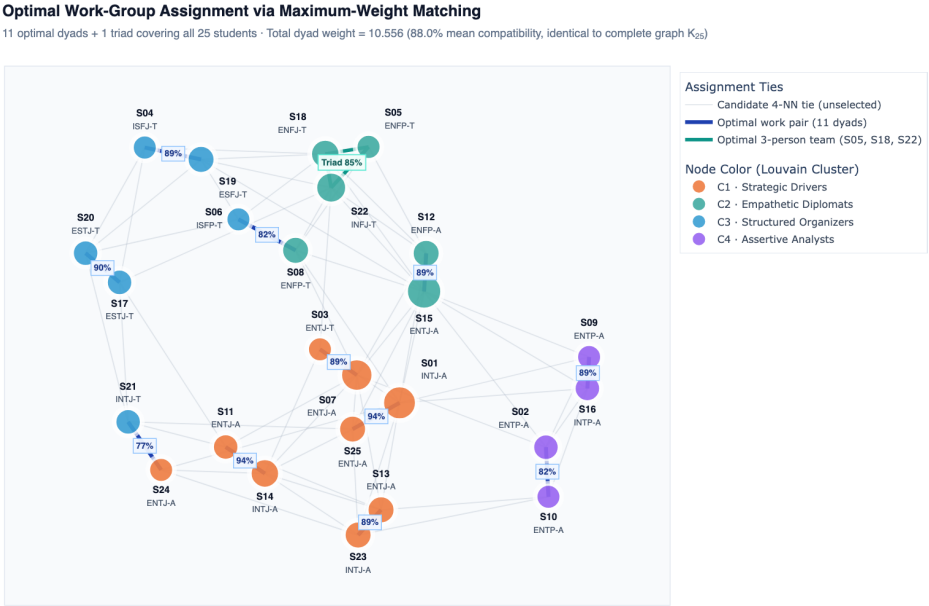


Figure 7: Optimal cohort allocation on G_{compat} (11 blue dyads and 1 teal triad), colored by Louvain community.

Finally, Table 8 summarizes hyperparameter sensitivity across $k \in \{3, 4, 5\}$ and $\alpha \in \{0.55, 0.65, 0.75\}$ (complementing our fine-grid $\alpha \in \{0.55, 0.60, 0.65, 0.70, 0.75\}$ check at $k = 4$, where edge Jaccard stays in $[0.896, 0.973]$ with 12/12 identical optimal dyads). Every configuration remains 100% connected, top-5

hub overlap holds at 4/5–5/5 on strength (3/5–5/5 on hop betweenness), and **all six settings with $k \in \{4, 5\}$ recover the exact same 12/12 optimal dyads** (88.0% fit; 10/12 at $k = 3$, 87.6%).

Table 8: Hyperparameter sensitivity around the reference model ($k = 4, \alpha = 0.65$).

k	Anchor α	Edges	Conn.	Weighted Q	Comm. ARI	Top-5 Str.	Top-5 Betw.	Shared Dyads (Fit)
3	0.55 / 0.65 / 0.75	56 / 56 / 57	Yes	0.415 / 0.393 / 0.406	0.583 / 0.583 / 0.360	4/5, 4/5, 5/5	3/5, 3/5, 4/5	10/12 (87.6%)
4	**0.55 / 0.65 / 0.75**	**73 / 73 / 75**	**Yes**	**0.377 / 0.340 / 0.324**	**0.613 / 1.000 / 1.000**	**4/5, 5/5, 5/5**	**4/5, 5/5, 5/5**	**12/12 (88.0%)**
5	0.55 / 0.65 / 0.75	90 / 91 / 90	Yes	0.348 / 0.348 / 0.324	0.613 / 0.613 / 0.292	4/5, 5/5, 5/5	4/5, 5/5, 4/5	12/12 (88.0%)

5 Conclusions

Modeling continuous *16Personalities* scores via a normalized L_2 target-partner 4-NN graph ($N = 25$) yields a compact, complementary team network ($r_E = -0.463$, 88.0% mean matched dyad fit) that refines discrete four-letter MBTI labels (ARI = 0.376), preserves max-min triad fairness (84.5%), and remains invariant in optimal dyad allocation across $\alpha \in [0.55, 0.75]$ and $k \in \{4, 5\}$. Simultaneously, our null-model and Spearman checks provide essential perspective: local hub prominence tracks centroid proximity ($\rho_s = 0.799$), and modularity $Q = 0.340$ reflects 5D k-NN spatial locality ($Q_{\text{perm}} = 0.351 \pm 0.038$) rather than multimodal gaps in raw traits. Future work should address non-response among the 8 missing classmates and calibrate trait weights empirically against post-project team outcomes.

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